Hospital Dashboard User Guide

Purpose of the Dashboard

This dashboard helps users compare hospital performance across states, ownership types, hospital types, and quality domains such as mortality, safety, readmission, patient experience, and timely care. It supports decision making by showing which hospitals are performing well and which areas need improvement.

Data Refresh

The dashboard is connected to hospital source data. Whenever new hospital records, ratings, or performance details are added into the source file, the dashboard can be refreshed to show the latest values.

How to Refresh Data

1. Open the report in Power BI Desktop.
2. Go to the Home tab.
3. Click Refresh.
4. Wait for the report to reload updated data.

Expected refresh time is around 2 to 3 minutes for approximately 5,000 rows of hospital data.

Slicers Used in the Dashboard

Slicers allow users to filter the dashboard and focus on specific hospital groups.

Common slicers available:

- State
- Hospital Ownership
- Hospital Type
- Emergency Services
- Birthing Friendly Designation
- Rating Threshold
- Measure Improvement Percentage

Example:
If a user selects California and Acute Care Hospitals, all charts and KPI cards update automatically for only those hospitals.

KPI Cards in the Dashboard

The dashboard includes KPI cards to quickly summarize important hospital performance information.

Main KPIs include:

- Total Hospitals
- Average Hospital Rating
- Percentage of Birthing Friendly Hospitals
- Better Count
- Worse Count

- No Different Count
- High Performing Hospitals
- Percentage of High Performing Hospitals
- Projected Better Percentage

These KPI cards automatically change when slicers are applied.

What If Sliders

The Scenario Analysis page includes What If sliders that allow users to test different conditions.

Rating Threshold Slider

This slider allows users to choose a rating value from 1 to 5.
The dashboard then recalculates how many hospitals qualify as high performing.

Example:

- Rating threshold = 3 → More hospitals qualify
- Rating threshold = 5 → Fewer hospitals qualify

Measure Improvement Slider

This slider allows users to simulate improvement percentages in hospital performance.

Example:

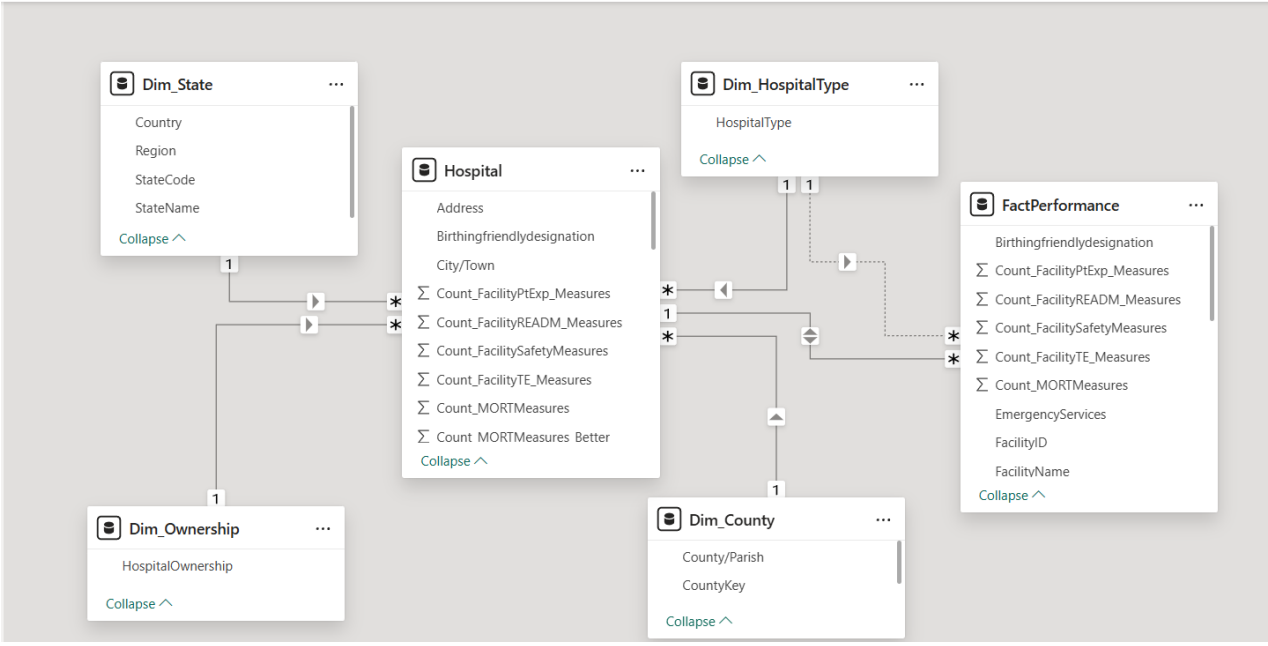
- Increasing the improvement percentage increases the projected percentage of better performing hospitals.
- This helps users understand future improvement scenarios.

Important Note

All visuals, KPI cards, and charts respond automatically to slicers and What If selections. This makes the dashboard interactive and useful for exploring different hospital performance scenarios.

Source information based on project documentation and dashboard build process.

Entity Relationship Diagram:



The data model follows a star schema structure, where the Hospital table acts as the central table connected to multiple dimension tables.

FactPerformance stores detailed hospital performance measures such as mortality, safety, readmission, timely care, and patient experience metrics.

Dimension tables such as Dim_State, Dim_County, Dim_HospitalType, and Dim_Ownership are used to support filtering, slicing, grouping, and detailed analysis across different hospital categories.

Publishing to Power BI Service requires a valid Microsoft account and Power BI access. Due to lack of login credentials, the dashboard was prepared for publishing but could not be deployed online. The .pbix file, ERD screenshot, dashboard screenshots, DAX documentation, and user guide are provided as final deliverables.

DAX Measures Added:

1. Avg_BetterMeasuresPerHospital = DIVIDE([BetterCount], [TotalHospitals],0)
2. AvgRating = AVERAGE(Hospital[Overallrating])
3. BetterCount =CALCULATE(SUM(FactPerformance[MeasureCount]),FactPerformance[MeasureStatus] = "Better")
4. HighPerformingDynamic = DIVIDE([BetterCount], [TotalMeasureCount], 0)

5. HighPerformingHospitals = VAR SelectedThreshold = SELECTEDVALUE(RatingThreshold[RatingThreshold],3) RETURN CALCULATE(DISTINCTCOUNT(Hospital[FacilityID]),FILTER(ALL(Hospital),Hospital[Overallrating]>=SelectedThreshold))
6. HospitalsRating_4orAbove = CALCULATE(DISTINCTCOUNT(FactPerformance[FacilityID]),FILTER(ALL(FactPerformance),[AvgRating] >= 4))
7. MeasureImprovement Value = SELECTEDVALUE('MeasureImprovement'[MeasureImprovement], 0)
8. MortalityBetter = CALCULATE([BetterCount], FactPerformance[MeasureDomain] = "Mortality")
9. NoDifferentCount = CALCULATE(SUM(FactPerformance[MeasureCount]), FactPerformance[MeasureStatus] = "NoDifferent")
10. PctBetter = DIVIDE([BetterCount], [TotalMeasureCount])
11. PctBirthing = DIVIDE(CALCULATE(COUNTROWS(Hospital), Hospital[Birthingfriendlydesignation] = "Yes"), [TotalHospitals])
12. PctHighPerforming = DIVIDE([HighPerformingHospitals],CALCULATE(DISTINCTCOUNT(Hospital[FacilityID]),ALL(Hospital)),0)
13. PctMortalityBetter = DIVIDE([MortalityBetter], CALCULATE([TotalMeasureCount], FactPerformance[MeasureDomain] = "Mortality"))
14. PctNoDifferent = DIVIDE([NoDifferentCount], [TotalMeasureCount])
15. PctReadmissionBetter = DIVIDE([ReadmissionBetter], CALCULATE([TotalMeasureCount], FactPerformance[MeasureDomain] = "Readmission"))
16. PctSafetyBetter = DIVIDE([SafetyBetter], CALCULATE([TotalMeasureCount], FactPerformance[MeasureDomain] = "Safety"))
17. PctWorse = DIVIDE([WorseCount], [TotalMeasureCount])
18. ProjectedBetterCount = [BetterCount] + ([WorseCount] * SELECTEDVALUE(MeasureImprovement[MeasureImprovement]) / 100)
19. ProjectedPctBetter = DIVIDE([ProjectedBetterCount], [TotalMeasureCount])
20. ProjectedWorseCount = [WorseCount] * (1 - SELECTEDVALUE(MeasureImprovement[MeasureImprovement]) / 100)

21. RatingThreshold Value = SELECTEDVALUE('RatingThreshold'[RatingThreshold], 4)
22. ReadmissionBetter = CALCULATE([BetterCount],FactPerformance[MeasureDomain] = "Readmission")
23. SafetyBetter = CALCULATE([BetterCount],FactPerformance[MeasureDomain] = "Safety")
24. Top 10 Hospitals = IF([Hospital Rank by Rating] <= 10,1, 0)
25. TotalHospitals = COUNT(Hospital[FacilityID])
26. TotalMeasureCount = SUM(FactPerformance[MeasureCount])
27. TotalStates = DISTINCTCOUNT(Dim_State[StateName])
28. WorseCount = CALCULATE(SUM(FactPerformance[MeasureCount]),FactPerformance[MeasureStatus] = "Worse")

Instructions for Adding New Performance Domains

Objective

New performance domains can be added to the hospital dashboard to make the analysis more detailed and useful. Some examples are patient satisfaction, infection control, emergency response, staff availability, financial efficiency, and surgery outcomes.

The main aim is to make sure that every new domain follows the same structure and workflow as the existing dashboard.

Step 1: Understand the Domain

Before adding a new domain, first understand what exactly it is measuring.

Things to identify:

- What the domain measures
- Why it is important
- Whether higher values are good or bad
- What KPI logic will be used
- Who will use this information

For example, Patient Satisfaction can measure the percentage of patients giving positive feedback. In this case, a higher percentage means better performance.

Step 2: Identify the Data Fields

Every new domain should have some basic fields so that it can fit properly into the existing dashboard.

Common fields required are:

- Hospital ID
- Hospital Name
- State
- Domain Name
- Measure Name
- Measure Value
- Performance Category
- Threshold or Benchmark
- Reporting Period

Example:

Hospital ID	**Hospital Name**	**State**	**Domain**	**Measure**	**Value**	**Threshold**	**Performance**
H001	ABC Hospital	Texas	Patient Satisfaction	Positive Feedback %	84	80	Better

These fields help in filtering, comparing, and creating visuals.

Step 3: Prepare the Data

Before loading the data into Power BI, it should be cleaned properly.

This includes:

- Removing duplicate records
- Handling missing values

- Standardizing hospital names and state names
- Making sure values are numeric
- Checking that Hospital IDs match with the existing hospital table
- Keeping percentage values in the same format
- Formatting the reporting period properly

If the data is not cleaned, it can create wrong calculations and incorrect visuals later.

Step 4: Create Relationships

The new table should be connected with the existing tables in the data model.

Recommended relationships:

- Hospital Table to New Domain Table as One to Many
- State Table to New Domain Table as One to Many

This helps the new domain work properly with existing slicers like state, hospital type, ownership, and emergency services.

Step 5: Create DAX Measures

After the data is loaded, DAX measures should be created for analysis.

Example measures:

1. Average Satisfaction = AVERAGE(PatientSatisfaction[Score])
2. Better Count = CALCULATE(COUNTROWS(PatientSatisfaction), PatientSatisfaction[Performance] = "Better")
3. Worse Count = CALCULATE(COUNTROWS(PatientSatisfaction), PatientSatisfaction[Performance] = "Worse")

Other useful measures can include:

- National Average
- State Average
- Best Performing Hospital
- Worst Performing Hospital

- Percentage Above Threshold
- Trend Over Time
- Rank by State

These measures help in comparing hospitals and understanding performance quickly.

Step 6: Add Visuals

Visuals should be selected based on the type of data.

Some useful visuals are:

- KPI Cards for average score, better count, and worse count
- Bar Charts for comparing hospitals
- Maps for state wise performance
- Trend Lines for monthly or yearly changes
- Matrix Tables for hospital level details
- Donut Charts for performance categories
- Scatter Plots for comparing one domain with another

For example:

- Patient Satisfaction vs Readmission Performance
- Infection Rate vs Mortality Performance

Step 7: Update Filters and Slicers

The new domain should also work with the existing filters in the report.

Important slicers may include:

- State
- Hospital Type
- Ownership
- Emergency Services
- Reporting Period

- Performance Category
- Domain Name

This keeps the report interactive and easy to use.

Step 8: Validate the Results

Before publishing the report, the new domain should be checked properly.

Validation can include:

- Comparing totals with source data
- Checking averages and percentages
- Making sure performance categories are correct
- Testing filters and slicers
- Checking for duplicate counts
- Comparing DAX results with manual calculations

This step is important because even a small mistake can affect the final dashboard.

Step 9: Document the Domain

Every new domain should be documented properly.

The documentation should mention:

- Purpose of the domain
- Data source used
- KPIs and DAX measures created
- Relationships added
- Visuals used
- Main business insights

This makes it easier to maintain the dashboard in the future.

Possible Future Domains

Some domains that can be added later are:

- Patient Satisfaction
- Infection Rates
- Emergency Wait Time
- Staff to Patient Ratio
- Financial Efficiency
- Bed Occupancy Rate
- Surgery Success Rate
- ICU Performance
- Average Length of Stay
- Insurance Claim Processing Time

Final Summary

Whenever a new domain is added, the same process should be followed: understand the domain, prepare the data, create relationships, build DAX measures, add visuals, test the results, and document everything.

This keeps the dashboard clean, consistent, and easy to expand in the future.